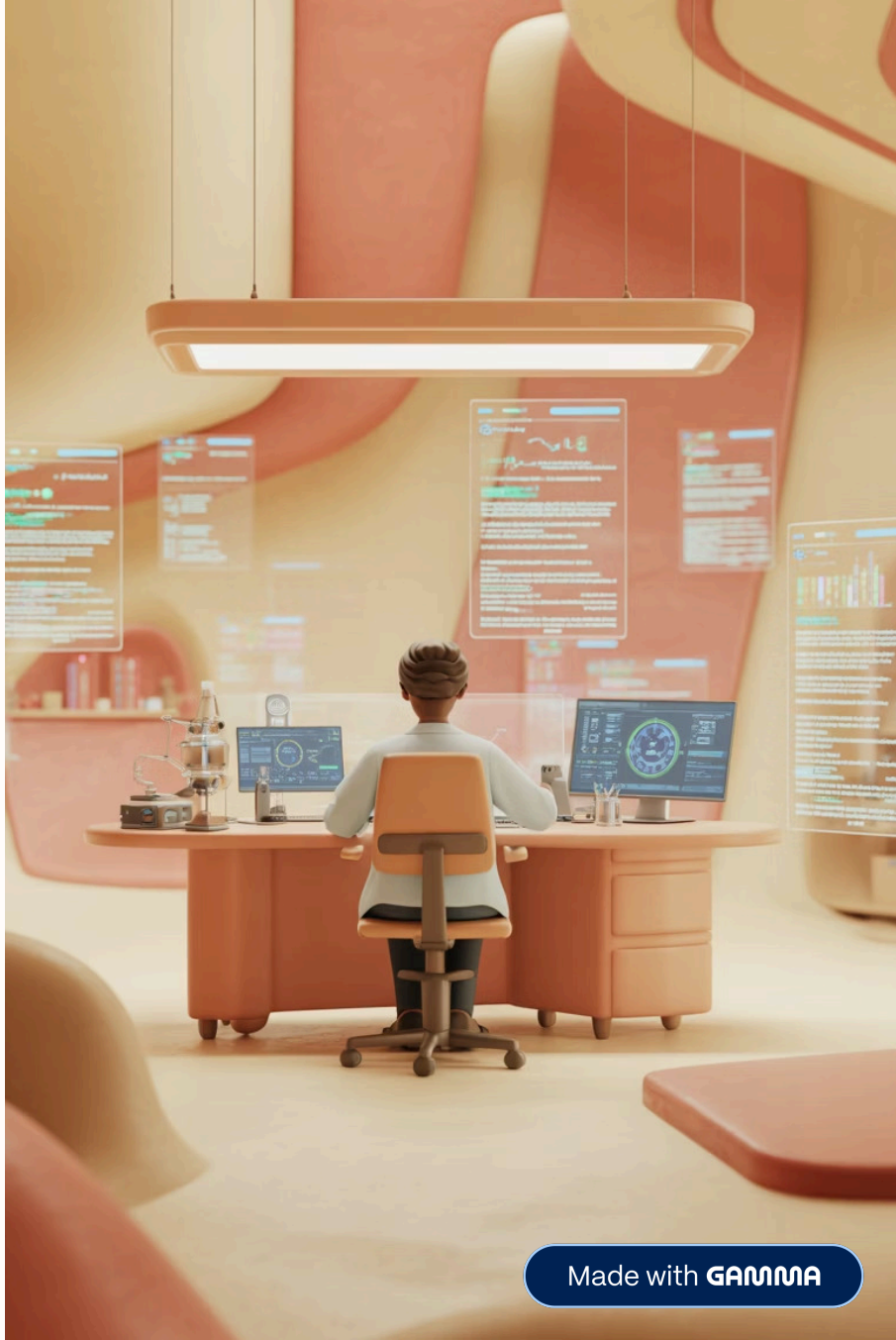
Academic Research Assistant

An MCP server connecting AI assistants to real academic papers — live from arXiv.



The Problem with Paper Search Today

🔍 Fragmented Workflow

Multiple websites, manual abstract reading, no unified interface

🤖 AI Hallucinations

Fabricated titles, outdated citations, unreliable references

⏱️ No Real-Time Data

Training data has a cutoff — AI can't access new research

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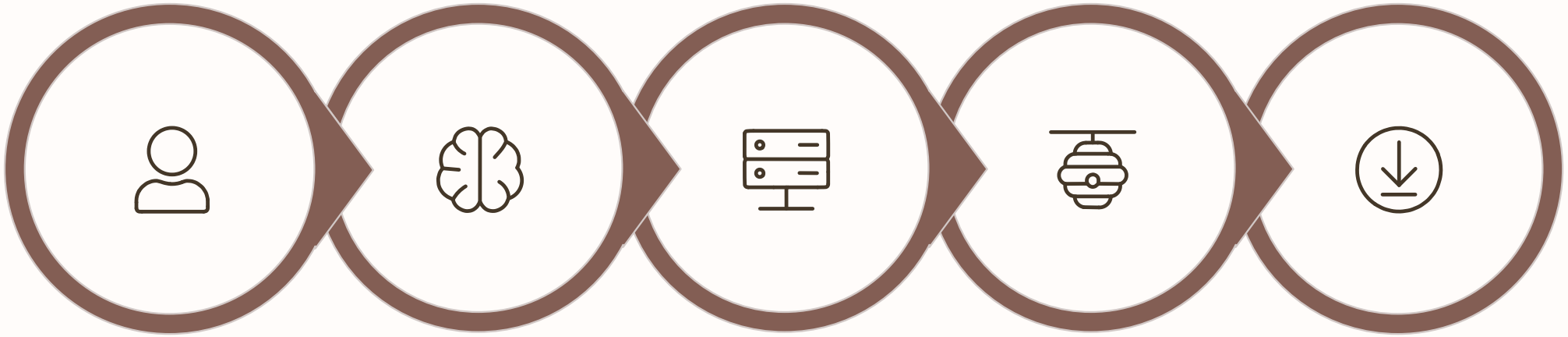
Our Solution

An MCP server that bridges the AI assistant directly to the arXiv API — live retrieval, no hallucinations.



User Question → **AI Selects Tool** → **MCP Executes** → **Query arXiv API** → **Return Results**

Every response is grounded in real, up-to-date scientific literature.

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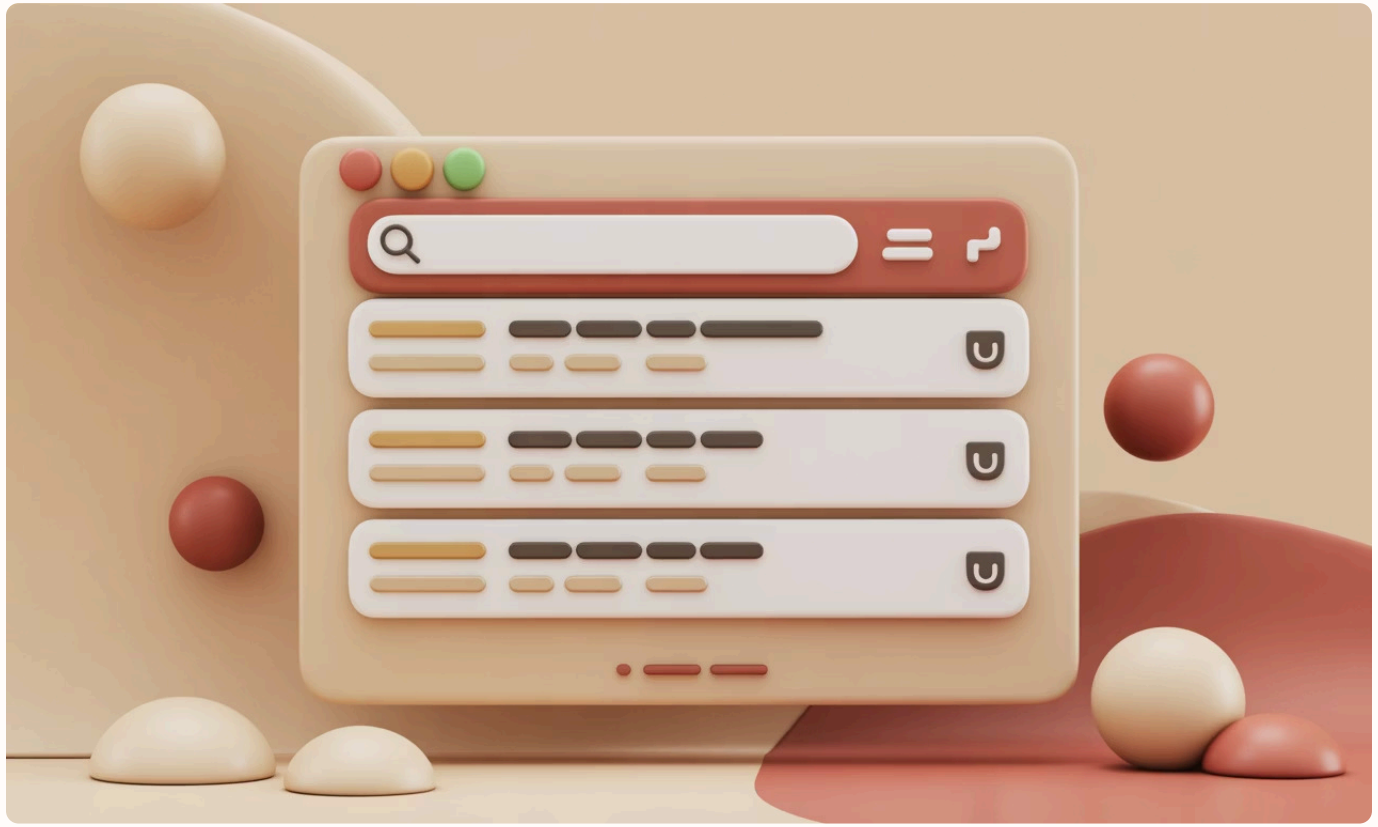

TOOL 1 + 2

Search & Inspect

search_papers — returns title, authors, date, categories, and PDF link for any topic.

get_paper_details — retrieves complete abstract and metadata for a specific arXiv ID.

Example: *"Get details for paper 2301.12345"*

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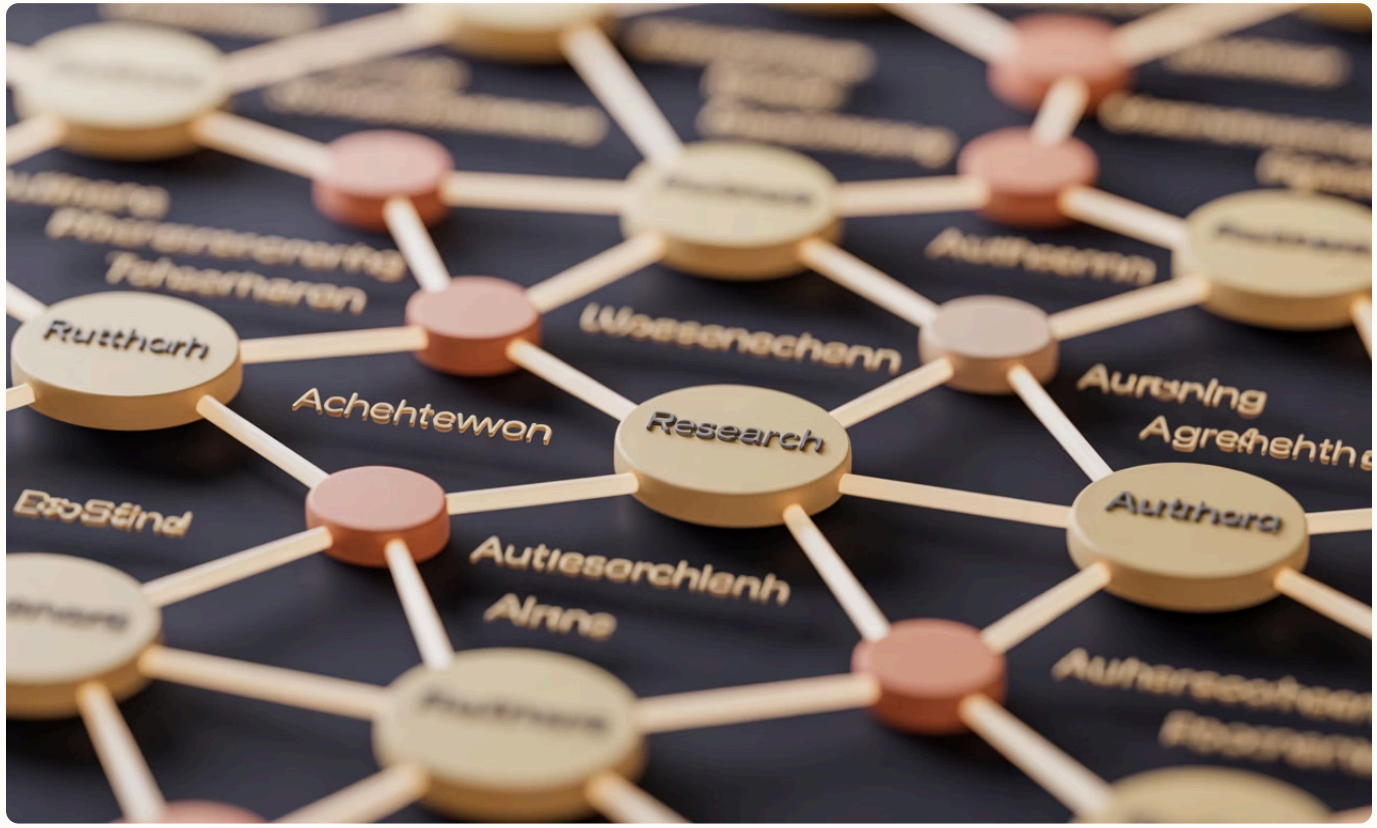
TOOL 3 + 4

Explore Authors & Related Work

search_by_author — finds all publications by a specific researcher.

get_related_papers — surfaces similar papers using shared research categories.

Example: *"What has Yann LeCun published recently?"*



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TOOL 5

compare_papers

Compare 2–3 papers side-by-side: title, authors, date, categories, and abstract summary — at a glance.

Similarities

Shared methods, overlapping categories

Differences

Distinct approaches, publication timelines

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Structured Prompting Strategy

This tells the AI how to solve research tasks step by step. Instead of randomly calling tools, it follows a clear workflow.

This multi-tool reasoning is the core of MCP systems.

Discovery	Deep Dive	Related Work	Synthesis
search_papers	get_paper_details	get_related_papers	compare_papers

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System Evaluation: With vs Without MCP

	Without MCP	With MCP
Data Source	Relies only on training data	Direct access to arXiv API
Paper Information	Hallucinated titles, outdated references, generic answers	Real titles, actual authors, current publication dates, PDF links
Workflow	Single tool calls, limited reasoning	Structured multi-step workflows: search → analyze → compare
Accuracy	Unreliable, may contain errors	Accurate, up-to-date, verifiable

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